Causality - SoSe 2025

Counterfactuals

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So Far

Throughout the lecture, we have touched the following topics:

- Observational Data vs. Randomized Controlled Trials
- Markov Properties and DAG Models (& d-Separation)
- Causal DAG Models and Causal Inference:
  - Parent-Adjustment
  - Path-Method
  - Back-Door and Front-Door Criterion
  - do-Calculus
- Causal Discovery:
  - Faithfulness & Causal Minimality
  - Constraint-Based CD (SGS, PC)
  - Score-Based CD (GES)
  - Restricted Models (LiNGAM, RESIT)
- Causal Machine Learning:
  - Out of distribution generalization (IV, Anchor)

Outline

In today's lecture, we will consider the last rung of Pearl's causal hierarchy and discuss

- Counterfactuals
- Q & A

Counterfactuals

Counterfactuals

What is a counterfactual?

A hypothetical scenario that lies in the past. We need our *imagination* to *reason* about counterfactuals. We *can never observe* them.

Example counterfactual questions:

- Would Kennedy be alive if Oswald had not killed him?
- Was it the aspirin that cured my headache?
  - (here we ask for a counterfactual explanation)

Counterfactuals

Interventions

Methods:

- backdoor adjustment, path method, IV regression, etc.

Questions:

- Address questions for the **population** or a **typical individual** (causal effect, average treatment effect, etc.)

Examples:

- "Smoking causes cancer"
- "ATE is 0.5" "

Counterfactuals

Methods:

- Abduction, Action, Prediction method (see later)

Questions:

- Personalized causation, explanation for individual

Example:

- "my uncle Joe, who smoked a pack a day for thirty years, would have been alive had he not smoked"

> Note that for the counterfactual question, it is not possible to compare with another person that also smoked a pack per day for thirty years.

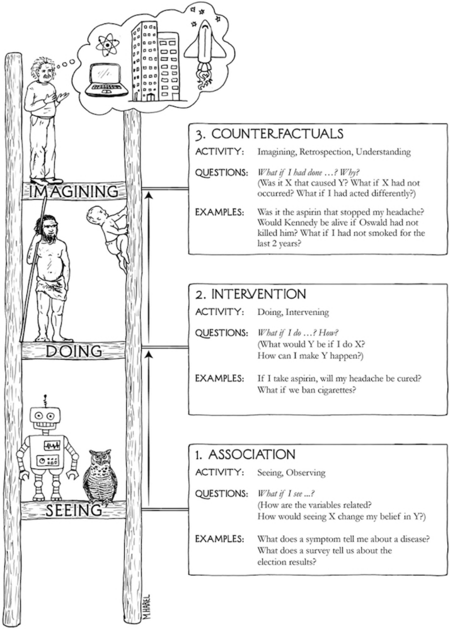

Figure 1: Ladder of Causation, J. Pearl, 2018

$p(y_{X=x} \mid Y = y', X = x')$

Requires the causal graph, and the structural causal model.

$p(y \mid do(X := x))$

Requires interventional data, or knowledge of the causal graph.

$p(y \mid X = x)$

No causal requirements.

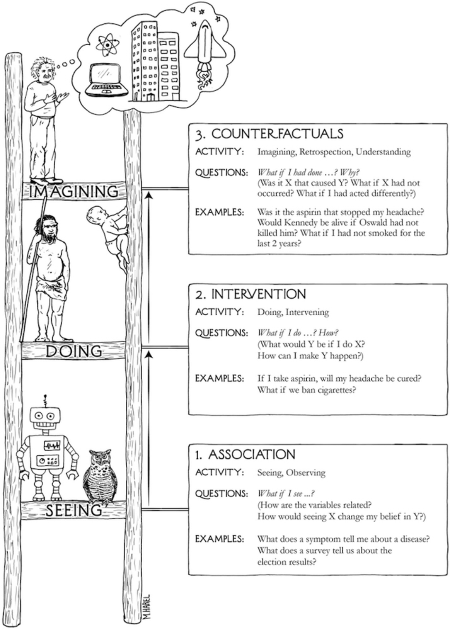

Figure 2: Ladder of Causation, J. Pearl, 2018

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Requires the causal graph, and the structural causal model.

$p(y \mid do(X := x))$

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$p(y \mid X = x)$

No causal requirements.

Counterfactuals - Notation

Definition Given an SCM $\mathcal{C} = (\boldsymbol{S}, P_{\boldsymbol{N}})$ which defines a set of random variables $\boldsymbol{X}$, where $\boldsymbol{X}$ contains Y and Z. The **counterfactual probability** is defined as[1]

$$p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{X}=\boldsymbol{x};do(Z:=z')}(y)$$

where $p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{X}=\boldsymbol{x}} = p_{\boldsymbol{N}|\boldsymbol{X}=\boldsymbol{x}}$, where the new noise variables need not be jointly independent anymore.

In words: "*Given that we* ***observed*** $\boldsymbol{X} = \boldsymbol{x} = (y, z, \dots)$, ***what would have been*** *the probability of* $Y = y$, *had we* ***set*** $Z = z'$ *by intervention?*"

[1] Note that this definition holds for the discrete case for the continuous case, we need to make some positivity assumptions (see ECI, Def. 6.17). On the previous slide we used shorthand notation $y_{X=x}$ for $y|do(X := x)$.

Example - Eye Disease

ECI Example 3.4: *For patients with some eye disease there is the choice between a treatment which sometimes cures the disease, but sometimes has bad side conditions.*

SCM:

$N_T \sim \text{coin}(\theta)$	$T := N_T$
$N_B \sim \text{coin}(0.01)$	$B := T \cdot N_B + (1 - T)(1 - N_B)$
$N_T =$ "treatment applied"	$T =$ "treatment applied"
$N_B =$ "unlucky group"	$B =$ "blind after one day"

Counterfactual question: *Assume a patient comes to hospital and gets treated ($T = 1$), but turns blind after a day ($B = 1$). Would the person be blind if he hadn't been treated?*

We can answer this query!

Example - Eye Disease (continued)

Initial SCM $\mathcal{C}$:

$$N_T \sim \text{coin}(\theta) \qquad T := N_T$$
$$N_B \sim \text{coin}(0.01) \qquad B := T \cdot N_B + (1-T)(1-N_B)$$

Query: Would the person be blind if he hadn't been treated?

Modified SCM $\mathcal{C} | T = 1, B = 1$:

- The data/observation $T = 1$ and $B = 1$ implies that $N_B = 1$

$$N_T \sim \text{coin}(\theta) \qquad T := N_T$$
$$N_B = 1 \qquad B := T \cdot N_B + (1-T)(1-N_B) = T$$

- Note: **only** modify the noise distributions!

Example - Eye Disease (continued)

Modified SCM $\mathcal{C}|T = 1, B = 1$:

$$
\begin{aligned}
&N_T \sim \text{coin}(\theta) \qquad && T := N_T \\
&N_B = 1 && B := TN_B + (1 - T)(1 - N_B) = T
\end{aligned}
$$

- Thus, the probability of not getting blind $B = 0$, given that we had chosen no-treatment ($T = 0$), however, knowing that for this patient the treatment ($T = 1$) led (past tense of 'lead') to blindness ($B = 1$):

$$\underbrace{p^{\mathcal{C}|T=1,B=1;do(T:=0)}(B = 0) = 1}_{\text{counterfactual}}$$

- If the doctor does not know N_B, i.e. in the initial model,

$$\underbrace{p^{\mathcal{C};do(T:=0)}(B = 0) = 0.01}_{\text{interventional}}$$

Counterfactuals in Thee Steps

Theorem (informal)[2] Given an SCM $\mathcal{C} = (\boldsymbol{S}, P_{\boldsymbol{N}})$ the conditional probability

$$p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{E}=\boldsymbol{e};do(A:=a)}(b)$$

of the counterfactual sentence "If it were $A = a$ then $B = b$", given the evidence $\boldsymbol{E} = \boldsymbol{e}$ can be evaluated using the following three steps:

1. **Abduction:** Update $p_{\boldsymbol{N}}^{\mathcal{C}}$ by **evidence $\boldsymbol{e}$** to get $p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{E}=\boldsymbol{e}}$
   - Note that this joint might not factorize anymore
2. **Action:** Modify $\mathcal{C}$ by the **intervention** $do(A := a)$, i.e., change the course of history by intervening
   - We obtain $\mathcal{C}|\boldsymbol{E} = \boldsymbol{e}; do(A := a)$
3. **Prediction:** Compute the probability $p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{E}=\boldsymbol{e};do(A:=a)}(b)$ in the modified counterfactual model

[2] For more details see Theorem 7.1.7 in Pearl's "Causality" book.

Example - Ordinal Data

Given the following SCM, where $N_X, N_Y, N_Z \overset{iid}{\sim} \mathcal{U}(\{-5, -4, \ldots, 4, 5\})$:

$$
\begin{aligned}
X &:= N_X \\
Y &:= X^2 + N_Y \\
Z &:= 2 \cdot Y + X + N_Z
\end{aligned}
$$

Question: We observe the datapoint $(1, 2, 4)$ and want to know: *What would Z have been, had X been 2?*

Abduction: We compute that $p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{E}=(1,2,4)}$, where $\boldsymbol{E} = (X, Y, Z)$ has point mass for $(N_X, N_Y, N_Z) = (1, 1, -1)$.

Action: We obtain $\mathcal{C}|\boldsymbol{E} = (1, 2, 4); do(X := 2)$ as:

$$
\begin{aligned}
X &:= 2 \\
Y &:= 2^2 + 1 \\
Z &:= 2 \cdot 5 + 2 + (-1)
\end{aligned}
$$

Example - Ordinal Data (continued)

Action: We obtain $\mathcal{C}|\boldsymbol{E} = (1, 2, 4); do(X := 2)$ as:

$$X := 2$$
$$Y := 2^2 + 1$$
$$Z := 2 \cdot 5 + 2 + (-1)$$

Prediction: From the above, we see that

$$p_{\boldsymbol{N}}^{\mathcal{C}|\boldsymbol{E}=(1,2,4);do(X:=2)}(z)$$

has point mass on 11!

Questions

Consider the same SCM, where $N_X, N_Y, N_Z \overset{iid}{\sim} \mathcal{U}(\{-5, -4, \ldots, 4, 5\})$:

$$X := N_X$$
$$Y := X^2 + N_Y$$
$$Z := 2 \cdot Y + X + N_Z$$

We observe $(2, 5, 10)$ as evidence $\boldsymbol{e}$.

- What are (N_X, N_Y, N_Z) for this evidence?
  - $(2, 1, -2)$
- What would Y have been, had X been 1?
  - 2
- What would Z have been, had Y been 2?
  - 4
- What would Z have been, had X been 1?
  - 3

Counterfactuals in Machine Learning

Active research areas in ML include:

- **XAI** (counterfactual explanations)
  - Would Albert's loan have been denied had his income been 50.000€ per year?
- **Fairness**
  - Had Alice gotten the job, had she been male?
- **Medical Decision-Making**
  - What would have been the recovery time of patient A had she been treated with drug B?

The bottom line is:

Counterfactual reasoning requires strong assumptions!

We need to know the full SCM!

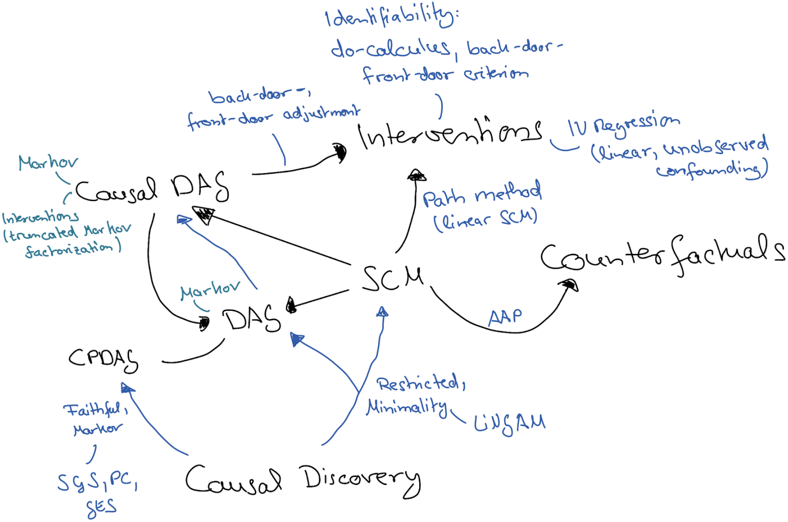

Figure 3: Mind map for this course. Disclaimer: Might be incomplete; arrows are not necessarily causal.

Credits

- **Recommended reading** ECI book Chapters 3.3 and 6.4, as well as (optionally) Chapter 8 of “The Book of Why” (Pearl, Mackenzie)
- This course *builds upon and extends* the material from *“Causality” (WiSe 2022/23)* that was kindly provided by **Stefan Harmeling**

Q & A

Thank you!